# EXPERIMENTAL ECONOMICS PROJECT

## ESPIONAGE GAME

Game designed by

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## Abstract

*In this paper, we have attempted to construct a simple variation of the Ultimatum Game by designing a fictional Espionage Game scenario where a spy and a government face-off against each other in an exchange of information for asylum. Conducted with two and a half dozen university students who were prepped for their respective roles of spy and government, our study found very little correlation with the Nash Equilibrium of the game. Instead, we got to know that there are other psychological factors that drove the minds of the players.*


## 1. Introduction

A spy (whistleblower) has been criminalized in his own country for misusing information relating to national security. He is thus now seeking asylum in another country in exchange for the significant confidential information he has of his own former country. He seeks to achieve this through a foreign (rival) country from which he asks for asylum and protection in return. This foreign government (country) wants the information against the first country and thus has to decide whether or not to grant asylum to the Spy in exchange for the information. These two interested parties are thus drawn into an imbroglio involving the exchange of asylum for covert information. Such is the premise of our game, the motivation for which has been largely drawn from trying to protract the classic Ultimatum Game to a more extended fanciful scenario.

The right of asylum (sometimes called political asylum) is an ancient juridical concept, under which a person persecuted by their own country may be protected by another sovereign authority or a foreign country. The most famous example of this scenario in recent times is the Edward Snowden fiasco wherein Edward Snowden, a computer professional, leaked classified information regarding mass surveillance, Government secrecy, and the balance between national security and information privacy to the media wherein he was charged for two counts of violating the Espionage Act and theft of Government property by the United States, forcing him to seek asylum in 21 countries. He was ultimately granted asylum and residence permit in Russia.

Our game seeks to construct a simplistic scenario of such a case and plot what the optimum reactions (read Nash Equilibrium) of the parties involved would be, given a full set of benefit-cost prescriptions and simplifying assumptions. We have tested the game on a small sample of university students, following proper experimental protocol, which has also been described in detail. We primarily aim to show statistically that the textbook results predicted by Nash Equilibrium are not as prevalent as popularly believed to be, thereby also showing the growing relevance and importance of Experimental Economics in the ever-growing jargon of Economic Game Theory. The results of our experiments and the underlying analysis and implications have been enumerated in probing detail in the pages that follow.

## 2. Economic Theory and Literature Review

Many economic situations where only two parties are trying to reach an agreement involve some form of bargaining. Such bargaining situations may often lead to an impasse with neither side

agreeing to the offer made by the other. In such situations, the party with the greater bargaining power may make a take-it-or-leave-it offer, essentially thereby serving an ultimatum. It is often the case that the other contracting party will accept this ultimatum offer from the stronger party and both sides are able to obtain their respective shares of the total pie. Alternatively, the other party may reject this ultimatum offer and can ensure that the stronger party also gets nothing. This is the behavior that is sought to be captured by the ultimatum game.

So, the Ultimatum Game is a type of game often played in economic experiments. The modus operandi of this game is as follows: Two players participate in the game. One of them is the Proposer and the other person is the Respondent, and a sum of money, say $100, is given to the proposer as an initial endowment. Now the Proposer is asked to propose a division of this amount between the two of them & the Respondent can either accept or reject the proposal. So if the Proposer chooses to give $x to the Respondent ($0 \leq x \leq 100$), the Proposer will get to keep $(100 – x). If rejection takes place then neither player gets anything but if the second player accepts, then the money is split according to the proposal. Note that the money is not distributed unless the Respondent accepts the offer. The basic purpose of playing an Ultimatum Game is to check the validity of theoretical predictions about the assumption of rationality in self-interest maximizing individuals.

The game is played under the assumption that both players are rational self-interested individuals and both know that the other player is also a self-interested payoff maximizing individual (common knowledge). Hence both players care only about their own payoffs. Under such axioms, the Proposer offers a Proposer-Respondent split of 99-01 and the Respondent accepts. If the Respondent did not accept, he would not get even the one dollar that he now gets as he accepts the offer. This is the theoretical construct of the ultimatum game.[1]

Werner Guth, Rolf Schmittberger & Bernd Schwarze, economists at University of Cologne first conducted an experiment on the Ultimatum Game by taking 42 post graduation students as subjects and their results were quite different from the actual theoretical prediction. First, there were very few offers giving a very low amount to the Respondent. Second, relatively more generous offers were also rejected. In other words, both Proposers and Respondents behave in a manner that bring into consideration factors other than pure self-interestedness. These other factors are explained in the form of Altruism and Fairness. We will focus on the fairness aspect since it has some bearing on the results in this paper as enumerated later.

Since then several studies have been conducted by a large number of economists working in this field. Anderson, Steffen, Seda Ertac and Uri Gneezy published their paper 'Stakes Matter in Ultimatum Game' in American Economic Review (volume 101; No 7, December 2011). Their main result was that proportionately equivalent offers are less likely to be rejected with high stakes compare to the case of low stakes. In fact, this paper basically tried to present that as the stakes increase, the rejection rate approaches zero.

The Social Ultimatum Game was conducted by three economists named Yu-Han Chang, Tomer Levinbimo & Raji Maheswaram. Hessel Oosterbeek, Randolph Sloof and Gus Van De Kulien

[1]Gautam Gupta, Labyrinths of the Mind Chapter 2

subsequently published their paper “Cultural difference in Ultimatum Game Experiment: Evidence from a Meta-Analysis” in Experimental Economics, June 2004, Volume 7, Issue 2, pp 171 - 188. This paper reports the finding of a meta-analysis of 37 papers with 75 results from Ultimatum Game experiments. As the results came from different countries, so they tried to investigate whether bargaining behavior in Ultimatum Game differs across the countries. They found differences in the behavior of respondents (and not of proposers) across geographical regions.

Roger Berger, Heiko Rauhat, Sandra Prade & Dirk Helbing worked on a different type of Ultimatum Game. Their paper was titled “Bargaining over time in Ultimatum Game Experiments”. Instead of money, here time was taken as the endowment. The subjects imposed waiting time on others (respondents). Here also the results were quite similar with the previous experimental results. This provides a brief description of some of the most well-known papers done on various types of experiments on Ultimatum Games that have taken place all over the world.

## 3. Backdrop of the Game

The Espionage Game (a form of an Ultimatum Game) on which we have conducted the experiment basically deals with a Spy (defector) who has a secret information about his own country and wishes to reveal it to another country (in the process extracting his own benefits).

In the practical world, there have been several infamous defectors or spies who have leaked confidential information to other countries, thus creating a situation of turmoil between countries. A number of studies, sociological, political, economic as well as psychological, have been made to get an idea about the behavior associated with the risks of espionage. Some of the studies did mention the following results:

- Selling secrets is seldom the result of a sudden, uncontrolled impulse. It is usually **the last act of a long-simmering emotional crisis**. In many cases, the symptoms of this crisis have been observable, identifiable, and even treatable before the damage was done. Typically, however, the potential significance of the "at-risk" behavior has not been recognized or reported at the time by co-workers or supervisor.
- Spies are not "crazy," but they usually are **emotionally disturbed** or suffer from one or more personality disorders. A personality disorder is recognizable as a pattern of behavior that is poorly adapted to the circumstances in which it occurs, leading to conflicts in relationships, difficulties at work, and periodic emotional shifts. Behavior can become self-defeating and sometimes self-destructive. Of the personality disorders found in spies, the two most common are antisocial personality disorder and narcissism. These two disorders have some characteristics in common and are sometimes found together.

A person with an **antisocial personality disorder** tends to reject the normal rules and standards of society. (Antisocial, in this sense, is a technical term in psychology. It has nothing to do with not being interested in making friends.) The hallmark of people with antisocial personality is a lack of any feelings of guilt or remorse when they do something wrong. The values that in most people inhibit illegal behaviour are lacking.

The hallmarks of a **narcissistic personality** are unwarranted feelings of self-importance or self-esteem (grandiosity), a sense of entitlement, and a lack of empathy for others. Many successful over-achievers have narcissistic tendencies, commonly known as a large ego. The need to live up to their own high self-image may be what drives them to be successful.

**The general motives that drive the mind of a Spy are:**

**Money**- This is a general category that would include such selfish motivations as avarice (extreme greed for wealth or material gain) as well as what might be considered more noble motives such as the need to pay for a family member's medical treatment or a child's education. In any event, the Spy comes to the personal conclusion that espionage is the best or perhaps only means of obtaining the money desired.

**Ideology -** In the case of espionage, a particular ideology may serve as either the actual motivation for a Spy to breach the trust placed in them or simply as a means of rationalizing that behaviour. The so-called Cambridge Five were likely "true believers" whose motivation for working with the Soviets against their native United Kingdom was based largely (but not exclusively) in a utopian belief in Communist ideology.

**Compromise / Coercion -** This is a negative rather than positive form of motivation and can be equated with what one might think of as "blackmail" or perhaps even torture. In this case the Spy does not act of his or her own free will but, rather, is effectively forced to commit espionage through fear of punishment, exposure of wrongdoing, or some other undesirable outcome.

**Ego -** This could be considered the all-inclusive category, since an individual's opinion of his or herself and the effort he or she puts into enhancing or defending that opinion is fundamental to their decision-making.

**Some of the Famous Defectors in History:**

**Richard Sorge -** He was a Soviet military intelligence officer, active before and during the Second World War, working under cover of being a German journalist in both Nazi Germany and the Empire of Japan. His codename was "Ramsay". Sorge is most famous for his service in Japan in 1940 and 1941, when he provided information about Adolf Hitler's plan to attack the Soviet Union, although he did not succeed in finding out the exact date of the attack.

**Igor Gouzenko -** He was a cipher clerk for the Soviet Embassy to Canada in Ottawa, Ontario. He defected on September 5, 1945 with 109 documents on Soviet espionage activities in the West. Gouzenko exposed Joseph Stalin's efforts to steal nuclear secrets, and the technique of planting sleeper agents. The "Gouzenko Affair" is often credited as a triggering event of the Cold War.

**Aldrich Ames -** He is an American, convicted in 1994 of espionage against his country. He is serving a life sentence without the possibility of parole in the high-security Allenwood U.S. Penitentiary. Ames was formerly a 31-year Central Intelligence Agencycounter intelligence officer and analyst who committed treason against his country by Spying for the Soviet Union and Russia.

**Edward Snowden -** An American Computer Professional and a former system administrator for the Central Intelligence Agency, he leaked classified information from the National Security Agency (NSA) stated in June 2013. Snowden's leaked documents revealed numerous Global Surveillance Programmes; many of them run by the NSA and Five Eyes with cooperation of Telecommunication companies and European Governments. Snowden has been variously called a "Hero", a "Whistleblower", "Dissident" and a "Traitor" due to such deeds.

**Cambridge Five -** They were the most famous and romanticized spy ring. The Cambridge spy ring loosely refers to a number of Soviet spies recruited from Cambridge University in England. While there is debate over how many members were a part of the ring, the ring is most known for five members, Kim Philby, Guy Burgess, Don MacLean, Anthony Blunt and John Cairncross. They caused severe damage to the United States and Britain, both in terms of the depth and value of the intelligence that was compromised and with devastating blows to the political and intelligence communities on both sides of the pond. They attained significant position within the British Government.

## 4. Design of the Game – Structure and Solution Strategy

We consider a two-country world with two players:

- A **Spy** (whistleblower) who has been criminalized in his own country for misusing information. He is thus now seeking asylum in another country in exchange for the significant confidential information he has of his own former country. He seeks to achieve this through a foreign (rival) country from which he asks for asylum and protection in return.
- A **Foreign Government** which wants the information against the Spy's former country and thus must decide whether or not to grant asylum to the Spy in exchange for the information.

In this variant of an ultimatum game, we model the players' actions purely on selfish interests and the ability to strategize over an infinite time horizon. The game can continue for infinite rounds because we assume that the Spy has a large stock of information from which he draws out information worth $100 in each round (this conversion to monetary units is assumed to act as an incentive to the players who being university students may find it difficult to fit into strategic roles of an internationally sought Spy or a person representing a Government) and has to decide how to split the figure so as to elicit acceptance from the foreign Government. The information held by the spy is assumed to be homogenous across all units.

Thus, the game is played over multiple rounds, with the Spy offering some fraction of the total units of information to the Government in each round: The Spy and the Government bargain with each other – **information versus asylum**. At each round, if the Government accepts the Spy's offer, the Spy gets a year of asylum and after the expiration of this period the game goes to the next round where the contract for another year is drawn up again depending on whether the Spy's offer is accepted in that round, and so on. Thus, this exchange can go on year after year, as long as the Spy has meaningful information to provide, and this feature essentially makes this an infinite period game. The game can end at any round, but we have played it over five rounds only.

The Spy's strategy is designed in the following way: he knows that the foreign Government values the information that he can provide and thus does not have any outright motivation to reject him. He also knows **that the Government will have to bear a cost to grant him asylum**, so that if he has to secure acceptance from the Government, he must offer him slightly more than the Government's cost of maintenance. Thus, the Spy's strategy will be to figure out an offer within the range [0,100] that will cover the Government's cost as well as leave him with the maximum amount possible, because this remaining amount will be his payoff from that round. The benefit from asylum has not been accounted explicitly in monetary terms, since the Spy is not given any money in return for the information, but only shelter. If the **Spy is rejected by the Government in any round, however, then the game ends there** and the Spy is arrested and tried in his former country.

Now, from the Government player's perspective, there are two actions in his strategy set: he can **either accept or reject the Spy, depending on the cost of granting asylum and the offer that the Spy makes**. However, we assume that the Government's cost does not remain fixed over all rounds; rather the cost can be high or low in any round which will be randomly determined for the Government player. The high cost corresponding to granting asylum is taken to be $50, whereas the low cost will be $20. At any round, only the Government player will know how much cost it has to bear (that is, high or low) but the Spy player will not know this. Thus, although the Spy knows that the Government faces two types of costs and also the respective magnitudes of them, he will not know what the Government's cost at any given round is.

An important point is that the real exchange of information occurs only after the Government accepts the Spy at any particular stage: thus, effectively the Spy will only make an offer at the beginning of the round and if the Government accepts the offer then the barter of information takes place. This means that if at any round the Government rejects the Spy's offer, the Government gets nothing in that round (or in any successive round because the game ends there) and the Spy also gets nothing, and is jailed in his country of origin.

The payoff structure is shown in the following extensive form diagram:

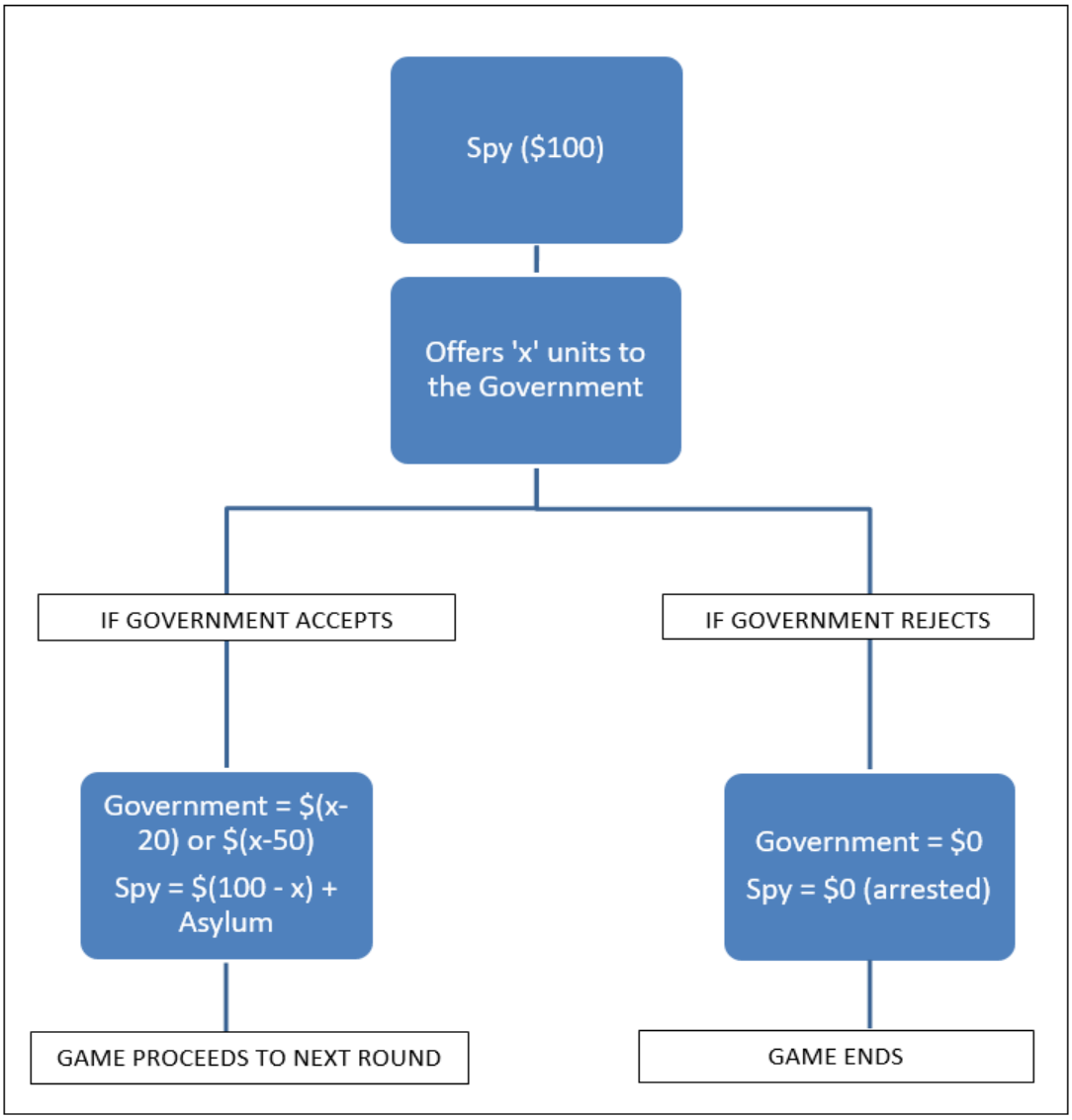


**Figure 1: Layout of the game**

## Equilibrium of the Game and Probable Payoffs:

It is evident from the game design that this offshoot of an ultimatum game has been modeled on incomplete information, unlike the original version of the game. Whereas in the original ultimatum game the Proposer can make an arbitrary offer which has to elicit an acceptance from the Respondent, neither player's **type** is explicitly considered there. Here, however, with the introduction of a cost distinction on part of the Government player (loosely representing the Respondent player) the **type** of this player will have a bearing on the path of play and ultimate solution strategy.

In this game of incomplete information, the Government player can be either **high cost type** or **low cost type** and this information is known to both players, along with the values assigned to the high and low costs. What the Spy does not know, however, is whether in any particular round the Government player will have a high cost or low cost. For example, here the high cost is assumed to

be $50 and low cost assumed to be $20; in any round the real cost will be known only to the Government player and not the Spy player. Thus, following Harsanyi's terminology, Nature (assumed to be another player not interested in the outcome of the game) reveals the Government's type to the Government player only and not to the Spy player: this is the incomplete information aspect of this game.

In its extensive form, the game is presented in the following figure:

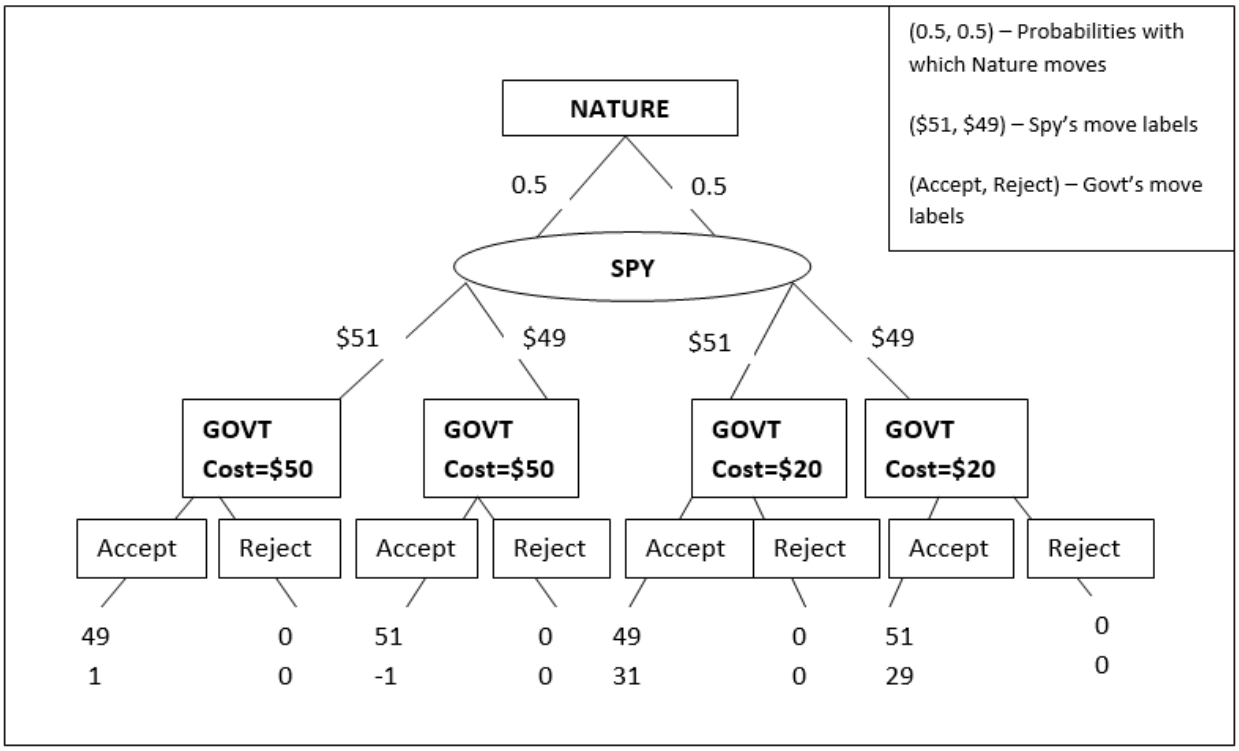


**Figure 2: Extensive Form of the Game (Specific Nash Equilibrium)**

Now, from the game tree above it can be seen that the Government can be high cost type with probability 0.5 and low cost type with probability 0.5 and the Spy has to make his offer without knowing the Government's exact type in that round. Since, by our assumption the Spy's need for asylum acts as the primary motive behind approaching the Government, a rational player will never offer less than $20 worth of information, since below that amount the Government's cost exceeds the benefit from the information and hence the Spy is sure to lose asylum. Thus, when the Government's cost of asylum is $20, its dominant strategy will be to accept any offer above $20 made by the Spy, and this is can be seen from the particular working example with shares (51,49) taken in Figure 2 as well.

Similarly, any offer above $50 worth of information by the Spy is sure to be accepted because then the benefit from the information will exceed the Government's cost and thus yield him sure positive payoffs. Thus, when the Spy offers $51 or more (as is seen from the first sub-game of Figure 2), the Government's dominant strategy again is to accept the offer. However, any offer below $50 will result in negative payoffs for the Government and thus are sure to be rejected. The only possible explanation for accepting any lower offer in the face of high cost can be the possibility of extracting

additional information in subsequent rounds by the Government, because if it rejects the Spy at that round, the game ends and no potential information will be available to it.

The range $20 < x < $50 is the most important range because any convex combination within this range is like a gamble to the Spy, because he may or may not get rejected depending on the Government's cost only. Using an expected payoff maximization format, a comparison can be made between the choices the Spy player opts for:

Any offer marginally above $50 (that is $51 in this case, since we assume integer shares) will yield the Spy an expected payoff of
E(51) = 0.5*(49) + 0.5*(49) = 49. (Now this is a sure payoff.)

Any offer marginally below $50 (say, $49) will yield the Spy an expected payoff of
E(49) = 0.5*(0) + 0.5*(51) = 25.5.

Now, as the Spy goes on trying to maximize his payoff, he would consider paying lesser to the Government; so as he shrinks the size of his offer his expected payoff is also going to rise. At the threshold level of $20 his expected payoff will be
E(20) = 0.5*(0) + 0.5*(80) = 40.

The expected value of this gamble at the threshold offer level too is less than the sure-shot payoff of $49.

This means that the Nash Equilibrium of this game is the offer (51, 49), that is, the Spy should offer $51 to the Government and the remaining $49 is his payoff from that round. The offer (50, 50) can be tricky because at the cost of $50 the Government will just break even and the indifference relation may not necessarily imply acceptance.

Thus, the Nash Equilibrium of the Game is:

| | | |
|---|---|---|
| **Spy** | **:** | **($51, $49)** |
| **Government** | **:** | **Accept the Spy** |

## 5. Set up of the Game

The above mentioned game was experimentally played using thirty university students as subjects, there being fifteen groups each comprising of two subjects. One subject played the role of a Spy and the other that of the Foreign Government.

Although the game was designed as an infinitely continuing game, it was played only over five rounds. The number of rounds was not disclosed to the subjects so that the possibility of backward induction could be ruled out. Each player was given a separate instruction sheet, and the instructions were verbally explained as well. (The instruction sheets have been attached in the appendix section of this paper.)

It was explained to the subjects that any kind of interaction was not permitted once the experiment started, since it could lead to serious contamination of data. A double-blind protocol was followed, and strict anonymity has been maintained while handling and analyzing the data as well.

Since monetary remunerations could not be arranged for the subjects that could act as incentives for the subjects, some refreshments were provided at the end of play.

## 6. Analysis

The following is a detailed enumeration of the observations of our experiment and the implications they entail:
Number of subjects: 30
Number of groups: 15
Number of rounds: 5

| | Round 1 | Round 2 | Round 3 | Round 4 | Round 5 |
|---|---|---|---|---|---|
| **Player 9** | 50 | 40 | 40 | 50 | 60 |
| **Player 11** | 20 | 45 | 45 | 70 | 70 |
| **Player 17** | 52 | 55 | 85 | 40 | NA |
| **Player 25** | 70 | 72 | 65 | 60 | NA |
| **Player 27** | 54 | 55 | 44 | 32 | NA |
| **Player 29** | 61 | 60 | 64 | 55 | NA |

**Table 1a: Spy's offers**

The above table represents the offers made by the spy players in each round (Rounds one to five). Here we have considered the successive offers of those spy players who survived atleast four rounds. Among these six players, only player 9 and player 11 played in the last round. *The figure below represents the same.*

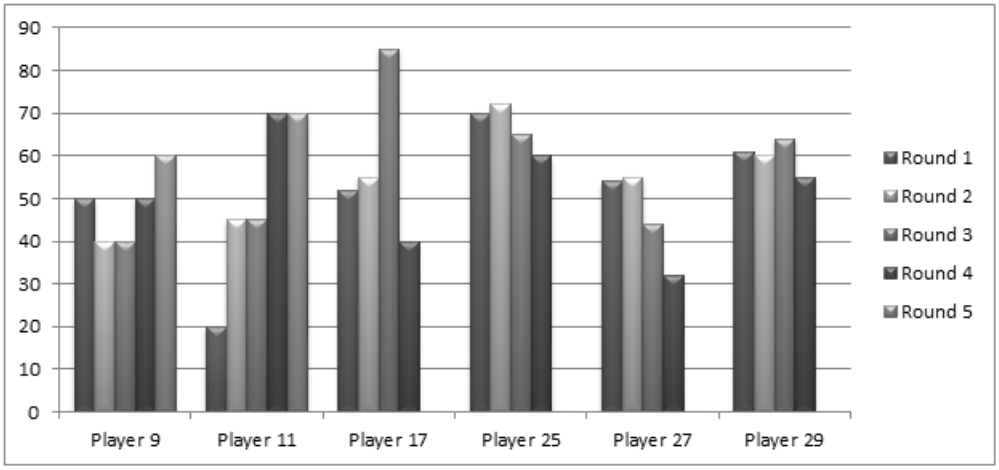


**Figure 3: Diagrammatic Representation of Spy's Offers**

We summarize the actions of these six spy players as follows:

**Player 9:** This player played quite consistently over the rounds.

**Player 11:** The offer made by this player increased in each successive round reflecting that the player could comprehend the game and play rationally as the game progressed. Another probable explanation for this successive increase in offers can be the fact that initially the player valued asylum less and thus played as a risk lover.

**Player 17:** This player played rationally till the third round and offered information amounts greater than 50. However, in the fourth round the player undertook a gamble by offering less than 50 and was rejected.

**Player 25 and Player 29:** These players played rationally throughout the game, as can be seen from the offers made by them in each round. However, they were both rejected by the Government in the fourth round. The possible motives guiding the Government Players' decisions have been discussed later.

**Player 27:** This player offered information amounts equal to or around the Nash offers in the initial rounds. However, his tendency towards risk increased as the game progressed. Finally, he was rejected by the Government in round four.

| Rounds | Government Cost=20 | | Government Cost=50 | |
|---|---|---|---|---|
| | Offer<=20 (In %) | Offer>20 (in %) | Offer<=50 (in %) | Offer>50 (in %) |
| 1 | 37.5 | 62.5 | 14.28 | 85.71 |
| 2 | 16.66 | 83.33 | 57.14 | 42.85 |
| 3 | 100 | 0 | 40 | 60 |
| 4 | 100 | 0 | 25 | 75 |
| 5 | NA | NA | 0 | 100 |

**Table 1b: Spy's offers**

The above table represents the Spy's offer to the Government for different cost levels in each round. Though the Spy does not know the costs, he makes an offer according to some beliefs. As per the assumptions of the game and the information available to the Spy Players, no rational Spy Player should offer an information amount less than 20 since such offers would be rejected by any rational Government Player. However, from the above table we find that 37.5% of the Spy Players offered an amount less than 20 in round 1, 16.6% in round 2 and 100% of the Spy Players in both rounds 3 and 4. The only possible explanation for this may be the fact that sufficient incentives could not be given to the Spy Players which would otherwise act as a Stimulus for these Players to take the game seriously. However, many others did play rationally as can be seen from the percentage of Spy Players offering an amount greater than 20.
Again, we find that for cost=50, 14.28% Spy Players offered an amount less than 50 in round 1, 57.14% of the Players offered an amount less than 50 in round 2. This figure dropped to 40%. Players in round 3 and 25% Players in round 4. This implies many Spy Players deviated from the Nash equilibrium (calculated as offer=51) of the game. This deviation from the Nash can be explained in the following manner:

These Spy Players are risk lovers because they work on the gamble that they may or may not get rejected depending on the Government's cost in that particular round. In case the Spy Player gets rejected, he loses the asylum and gets zero payoffs. However, if the Government turns out to have a low cost in a round, the Spy Player is bound to get more payoffs than the Nash equilibrium payoff. For example, if the Spy Player offers an amount of, say 21, and the cost of the Government turns out to be 20, then the Spy Player would get a payoff of (100-21) =79 and the Government would receive a payoff of 1. In addition, the Spy Player would also get his asylum. However, if the Spy Player values the asylum more than any information payoffs, he would always offer an amount greater than 50. This can be seen from the last column which depicts the percentage of Spy Players who offered an amount greater than 50.

| Rounds | Cost=20 | | | | Cost=50 | | | |
|---|---|---|---|---|---|---|---|---|
| | Offer<=20 | | Offer>20 | | Offer<=50 | | Offer>50 | |
| | Accept (in%) | Reject (in %) | Accept (in %) | Reject (in %) | Accept (in %) | Reject (in %) | Accept (in %) | Reject (in %) |
| 1 | 33.33 | 66.66 | 80 | 20 | 100 | 0 | 100 | 0 |
| 2 | 0 | 100 | 60 | 40 | 25 | 75 | 75 | 25 |
| 3 | NA | NA | 100 | 0 | 50 | 50 | 100 | 0 |
| 4 | NA | NA | 50 | 50 | 100 | 0 | 33.33 | 66.66 |
| 5 | NA | NA | NA | NA | NA | NA | 100 | 0 |

**Table 2: Government's Decisions**

The above table represents the Government's decision (accept or reject) in each round. We find for cost =20, the Government players reject the offer > 20 in 20% cases in round one, 40% cases in round two and 50% cases in round four. Similarly, for cost=50, we see that the Government players reject an offer > 50 in 25% cases in round two and 66.66% cases in round four. This reflects that the Government players have a tendency to reject an offer greater than the cost. This is because the Government perceives the offer as unfair representing bad behavior on the part of the Spy. Being prone to fairness the Government players respond by rejecting the offer. Though the Government gets zero payoff by rejecting the offer, this disutility is more than compensated for from the utility brought on from the satisfaction of seeing the Spy lose the asylum.

In Round 1, when the cost is 50 and the offer < 50, the offer has been accepted by the Government Players in 100% cases. Similarly, Government Players have accepted offers (<50) when the cost is equal to 50 in 25% cases in Round 2,50% cases in Round 3 and 100% cases in Round 4. A possible explanation for accepting any offer lower than the cost in that particular round can be the possibility of extracting additional information in subsequent rounds by the Government, because if it rejects the Spy at that round, the Government could lose out on potential information that he could have availed otherwise, in subsequent rounds. Remember that the players had no information as to when the game would end.

| Rounds | Percentage of Risk Lovers |
|---|---|
| 1 | 40 |
| 2 | 53.84 |
| 3 | 57.14 |
| 4 | 33.33 |
| 5 | 0 |

**Table 3: Percentage of Risk Lovers among Spy players**

The above table represents the percentage of risk lovers (among the Spy players) in each round. The percentage of risk lovers in each round is calculated by taking the percentage of Spy players offering an amount greater than 20 and less than 50 in each round. Since the Spy players were aware of the two cost levels incurred by the Government for providing the asylum, any offer lesser than 50 by the Spy player would imply risk loving behavior on his part. Ignoring rounds four and five (due to small sample bias) we find that the percentage of risk lovers increase in each round as the game progresses.

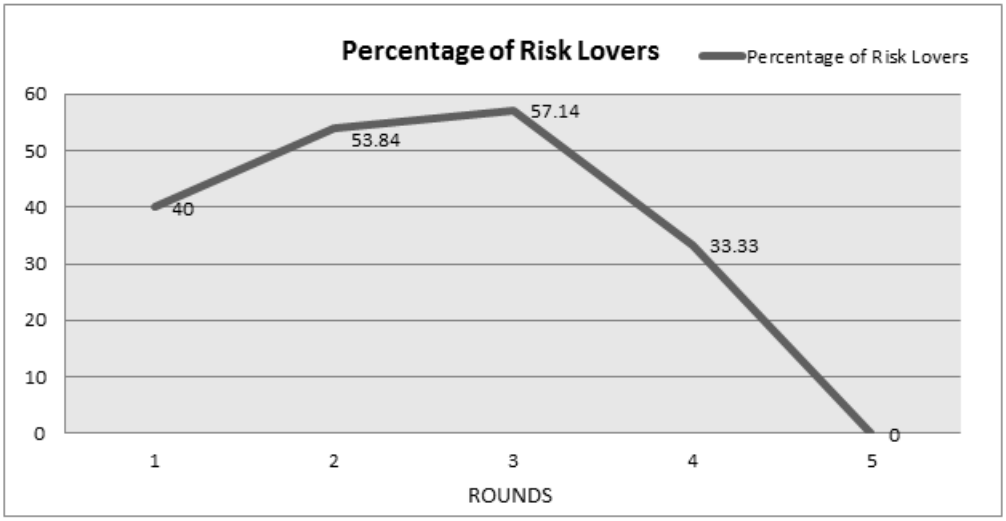


**Figure 4: Percentage of Risk Lovers**

In light of these results, we may perhaps infer that when the spy players see their offers (which are in the range $20 to $50) being accepted by the government player round after round, they begin to realize that the government player keeps accepting their offers either in expectation of a positive payoff in the next round or simply because the costs are $20, in which case their gamble is paying off. However, the decline after the fourth round indicates that government players now realize that they are being played for and the spy player will probably never make an offer above $50. In retaliation, the government player finally rejects their offer and ends the game, thus ending chances of the spy winning any further payoffs. By common knowledge, the spy player realizes he cannot fool the government player for long and reduces his risk-taking behavior in the remaining successive rounds.

## 7. Conclusion

The experiment that we have conducted has generated some mixed results. There have been situations where we have seen that the spy has made offers between $20 and $50. This shows their attitude towards risk (risk lovers). But their gamble tends to decline in the subsequent stages as they realized that the probability of the government accepting their offers would be low. There have also been instances of the spy offering amounts close to the Nash equilibrium value ($51) or more than that. In this regard, they played rationally where making the asylum deal was the prime motive. The deviations from the Nash equilibrium are clearly observed. These may be due to the following reasons: -

1. There is no desire for the Asylum or doesn't want to part with the information. (Offer less than 20)
2. Risk Loving attitude (Offer between $20 and $50)
3. Getting the Asylum is the sole purpose. (Offer above or equal to $50)

On the other hand, the theory predicts that any offer above $50 must be accepted on the part of the Government, and offers between $20 and $50 will be only accepted when the cost to the Government is $20 and finally offer less than $20 must be rejected. But we see some stark contrasts with what the theory has to say. There have been instances where the Government has accepted offers which are less than $20 and offers which were in-between $20 and $50 when the corresponding cost was $50. The Government also rejected offers above $20 (when the cost was $20) and offers above $50 (when the cost was $50). Again, these deviations may be explained as follows: -

1. The Government player wanted to extract some more information from the spy in future consecutive rounds or some kind of altruistic behavior (getting the asylum is perhaps necessary for the spy) guided him to accept any offers less than $20 and offers less than $50 (with the cost being $50).
2. The Government player rejecting offers above $50 is hard to explain. It may be due to the improper subjects that we have considered in this experiment, creating the small sample bias.

Thus, to conclude we see that theory predictions doesn't have much implications in the real world. As the behavior suggests, there has been significant deviations from the Nash outcome. There are different psychological factors which play in the minds of the people that stop them from being rational.

## 8. Limitations of the Experiment

Conducting an experiment is never an easy task. As Professor Gautam Gupta writes in his book “A Primer for Experiments in Economics”, Chapter 6 – the Methodology and Protocol of Experimental Economics, “It bears repetition that the design must be checked and re-checked, again and again”, there are several factors one needs to take into account before finally inviting subjects to be a part of the actual experiment. The experiment described in this paper was conducted by post graduate students of economics as part of their semester curriculum and most crucially, it was the first time they were running a full-scale experiment. To describe the limitations faced should not be construed as a façade for excuses, but rather as challenges that the team took upon themselves to overcome.

1. **Choosing the venue** – The experiment was conducted in a computer laboratory where one server was connected to 16 computers. This limited the number of subjects per round to 16, and to increase the sample size, more subjects could be included only after each play through of the game was completed.

2. **Monetary Incentives for the Experiment –** Monetary arrangements could not be provided to the subjects in the manner described in the game. Instead they were provided refreshments at the end of completion of the entire game without discriminating between spy players and government players all of whom received equal portions of the consumables distributed. Since the players knew this beforehand (not announcing any remuneration would have meant little or no volunteers) their incentive to earn bigger profits in the game might be reduced knowing that they would all receive the same packages. Participants, who were mostly economics students were thus requested to play in the spirit of the game and to play as rationally as possible.

3. **Simplified Scenario of the Game –** This has been the most serious drawback of our experiment. A critic will definitely point out that the assumption of homogeneity of secret information held by the spy is an unrealistic assumption. Given the nature of spies it is most unlikely that any of them will not think of duping the government with falsified information or attempt to double-cross them in some way. The assumption of infinite endowment for the spy is also a highly unrealistic assumption. The following are some modifications that could not be included in the final format of our espionage game:

   **a)** <u>There is a probability 'p' that the information offered by the spy is false.</u> The government has to take this probability into account when making his decision to accept or reject the spy's offer. Likewise, there can be a probability 'p' that the spy will be caught giving false information to the government after the offer has been made. This will force the spy to rethink his decision to offer false information on purpose.

   **b)** <u>The spy has a fixed endowment of 100units of information throughout the game.</u> No spy could possibly have an infinite endowment that lasts no matter how many rounds are played. Instead, he might have an endowment of 100units for the entire game and has to decide how much of it to offer to the government in each round. His payoff will be in the form of some remuneration (other than asylum) from the government while the government would accrue the benefit of the valuable information as his payoff.

4. **Motivating the subjects –** The espionage game was meant to analyze the choices made by the players purely against the basis of obtaining asylum and obtaining unique covert information, for the spy and the government respectively. This would enable us to obtain observations which are as close as possible to a practical real-life espionage scenario. However, university student subjects may not understand what asylum really means to an exiled spy or what confidential information means to a government. Thus, it was necessary to transform the language of the game into a maximizing-of-payoff scenario so that subjects would feel motivated by the thought of earning some cash or kind if not by asylum years and units of information. This metamorphosed the game into something it was not originally meant to be.

## 9. References

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3. Charney, D.L. and Irwin J.A. (2014). A Guide to the Psychology of Espionage. AFIO.

## 10. Acknowledgement

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Lastly, since this was a team effort, we are indebted to each and every one of our own team members, because this endeavor has been made possible only through the collective and simultaneous contribution of the individual members.

## 11. Appendix

**Instruction Sheet - SPY PLAYER**

- You are a Spy who has been criminalized in his own country for misusing information. You are thus now seeking asylum in another country in exchange for the significant confidential and valuable information you have of your own former country.
- You will be paired against a player representing a Foreign Government who will decide whether to accept your offer of information and hence provide you asylum, or reject it.
- The game will be played over multiple rounds, each round metaphorically equivalent to a period of one year. At each round, you will get an endowment of $100 worth of information from your total stock of information and you will decide how much of that to give to the Government. The highest offer you can make is 100 (i.e., the total endowment at that round) and the lowest you can

offer is 0, in integer amounts. The Government will get this information only after it has granted you the asylum and not before that.

- The government's decision will depend on the *amount of information you offer* and *his cost of providing you asylum*. The contract will be drawn up for one year if you are accepted, after the expiration of which you will approach him/her again, offering to reveal information in return for asylum. If, however, you are rejected at any round, then you will be arrested and tried in your own country and the game will end. The payoffs to both of you will be zero.
- The amount remaining with you after making the offer will be your net payoff at the end of each round, and the accumulated payoffs over all the rounds is your profit. e.g. If you offer $60 in one round and $50 in another round, suppose after which the game ends, you get $(100-60) + $(100-50) = $90 as your aggregate payoff.
- Before making the offer, you must remember that Government has to bear a cost for granting asylum to you and the cost may be high or low. The high cost of providing asylum is $50 and the low cost is $20 – this cost will be known only to the government in each round.
- Lastly, you need to note that *the game can end at any round* and you will not be told at the beginning when the game ends**.**

**Instruction Sheet - GOVERNMENT PLAYER**

- Suppose you represent the government of a country in a two-country world.
- A spy has significant valuable information that he wants to trade with you in exchange for asylum. You will play the game against the spy over multiple rounds, deciding whether to accept or reject the spy's offers.
- Each round will be metaphorically equivalent to a period of one year for which the contract of granting asylum will be drawn up. At each round, you can either *accept or reject the Spy's offer* of information depending on the *cost that you have to bear for granting asylum* and the *amount of information that the Spy reveals*. Note that by rejecting the Spy at any particular round, you will also lose out potential information that you could have gained at later rounds.
- At each round, the Spy will decide how many dollars' worth of information to reveal from his endowment of $100 specific for that round only.
- In each round the cost you have to bear can be either high or low with a 50-50 chance. The high cost is $50 and the low cost is $20, and these figures will be randomly generated for you. The Spy player knows the costs, but at any round he will not know whether you are incurring the high cost of $50 or the low cost of $20.
- The amount that the Spy offers to you will be your payoff for that round, and it can be any integer amount between 0 and 100, both inclusive. The offer that the Spy makes will be realized only if you accept the Spy's request, so that if you reject him/her at any round then the payoff at that round for you will be 0. The accumulated payoff over all the rounds will be your net profit from playing the game. E.g. suppose you get $60 in one round and $40 in the next round then your aggregate payoff over the two rounds is $(60+40) = $100.
- Lastly, you need to note that *the game can end at any round* and you will not be told at the beginning when the game ends.

## Z-TREE Programming

The following is a complete representation of the algorithm used in implementing our game in the Z-TREE programming software:

**Background Parameters:**

Subjects: 30
Groups: 15
Periods: 5

**Program:**

```
inf_end=100;
Participate=if(Type==2,1,0);
r=random();
cost_asy=if(r<0.5,50,20);

Label: The cost of asylum is
Variable: cost_asy
Layout:1
```

**Spy's Offer Stage:**

```
Participate=if(Type==1,1,0);

Label: Your initial endowment is
Variable: inf_end
Layout: 1
Label: Your offer to government
Variable: offer
Layout: 1
Input: Max: 100
       Min: 0
```

**Government's Acceptance Stage:**

```
Participate=if(Type==2,1,0);
offer=find(same(Group)&Type==1, offer);

Label: You have been offered
Variable: offer
Layout: 1

Label: Do you accept the offer
Variable: accept
```

Layout: !radio:
1=”accept”;
0=”reject”;
Input: Max: 1
Min: 0

**Spy’s Profit Calculation Stage:**

Accept=find(same(Group)&Type==2,accept);
Participate=if(accept==1&Type==1,1,0);
share=if(Type==1,inf_end-offer,offer-cost_asy);
Profit=if(accept==1,share,0);

Label: Your offer to government was
Variable: offer
Layout: 1

Label: Your profit in this stage
Variable: Profit
Layout: 1

**Government’s Profit Calculation Stage:**

Participate=if(Type==2,1,0);

Label: You were offered
Variable: offer
Layout: 1

Label: Your profit in this stage
Variable: Profit
Layout: 1